StablePFN: Stable Prediction with Causal-Aware Tabular Foundation Model

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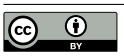
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Abstract

Pre-trained tabular prediction models based on Prior-Data Fitted Networks (PFNs), such as TabPFN and LimiX, have achieved remarkable progress in supervised learning, demonstrating immense potential across real-world scenarios and diverse downstream tasks. However, a critical question remains systematically unexplored: Does pre-training on data generated via causal mechanisms truly endow models with the ability to comprehend underlying causal structures? Furthermore, can these models leverage such causal information to achieve stable prediction across environments? To address these fundamental questions, we propose StablePFN, a novel tabular foundation model that integrates explicit causal awareness with stable predictive modeling. Leveraging a key yet largely overlooked advantage of the PFN paradigm, the availability of ground-truth causal structure during synthetic data generation, we train StablePFN to jointly identify the Markov Boundary (MB) of the target variable and perform the primary prediction task. We introduce an end-to-end "Decouple-Discover-Predict" architecture that utilizes sample decoupling weights to guide MB discovery, and employs a hard attention masking mechanism during inference to incorporate causal structural knowledge. Extensive experiments on both synthetic and real-world benchmarks demonstrate that StablePFN significantly outperforms state-of-the-art baselines in cross-environment prediction settings, particularly in challenging high-bias scenarios.

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CCS Concepts

• **Computing methodologies** → **Neural networks**; *Causal reasoning and diagnostics.*

Keywords

Stable Prediction, Tabular Foundation Models, Prior-Data Fitted Networks, Markov Boundary, Causal Discovery

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1 Introduction

Tabular data represents a ubiquitous and commercially significant data format in machine learning and data mining. For years, the field of tabular learning has been dominated by tree ensemble models [5, 13, 19, 32]. However, the emergence of Prior-Data Fitted Networks (PFNs) [27] has initiated a paradigm shift. By pre-training on vast amounts of synthetic data, PFN-based tabular models have demonstrated remarkable out-of-the-box predictive capabilities [12, 17, 18, 50]. The core philosophy of PFNs lies in using neural networks to approximate non-parametric Bayesian inference [28]. By exposing the model to diverse causal mechanisms during the pre-training phase, PFNs can perform effective In-Context Learning (ICL) on unseen real-world tabular tasks, showing immense potential across various downstream applications [25, 33].

Despite the fact that causal-mechanism-based data synthesis is a cornerstone of PFN success, a critical gap remains: Does the model truly understand the underlying causal structure? Existing PFN models are largely treated as powerful learners of posterior predictive distributions. Their black-box nature makes it difficult to determine whether the model has captured the causal relationships of the underlying data-generating process or is merely exploiting superficial statistical regularities.

This lack of mechanistic understanding becomes particularly detrimental in cross-environment scenarios. In real-world tabular applications, data distributions often shift over time or across different environments. Consequently, statistical correlations present in the training data may no longer hold in the test environment. Existing PFN architectures typically employ global attention mechanisms to process all features when outputting the posterior predictive distribution, which can lead the model to rely on spurious correlations, resulting in generalization failure in unknown test environments.

Therefore, an ideal tabular foundation model should not be a passive distribution estimator. Instead, it should possess the ability to actively identify and comprehend causal structures. In fact, the PFN pre-training paradigm inherently involves transparent causal data-generation mechanisms. However, most existing works utilize only the observational data generated by Structural Causal Models (SCMs), while neglecting the readily available causal structures inherent in the generation process.

To bridge the gap between powerful distribution fitting capabilities and mechanistic causal understanding, we propose StablePFN[1], a novel tabular foundation model that integrates Markov Boundary (MB) discovery with stable prediction. Unlike traditional PFNs that focus solely on predictive accuracy, StablePFN employs a multi-task learning framework designed to achieve sample decoupling, structure discovery, and stable prediction simultaneously. Specifically, we design a three-branch "Decouple-Discover-Predict" architecture. First, the model generates sample decoupling weights that facilitate the accurate discovery of the target variable's MB. Subsequently, we employ a hard attention masking mechanism during the inference phase to filter out non-causal features, forcing the model to rely exclusively on the identified MB to achieve stable predictive performance across environments. By pre-training on a massive scale of synthetic causal graphs, StablePFN acquires a generalizable structural recognition capability. This enables zero-shot stable prediction on unseen real-world datasets. Our main contributions are summarized as follows:

- We introduce a novel causal-aware pre-training paradigm for tabular learning that goes beyond pure posterior predictive fitting. By exploiting the inherent but underutilized structural transparency in PFN data generation, we bridge the fundamental gap between the powerful distribution fitting capabilities of PFNs and the necessity for mechanistic causal understanding.
- We propose StablePFN, an end-to-end framework built around a "Decouple-Discover-Predict" pipeline. Unlike previous black-box models, we guide MB discovery through sample decoupling, and then inject structural knowledge into the prediction branch via hard MB masking on the attention mechanism during inference to achieve stable prediction.
- Extensive experiments on diverse synthetic and real-world benchmarks demonstrate that StablePFN significantly outperforms state-of-the-art baselines, including the recent TabPFN v2.5 [12] and LimiX [50], under cross-environment settings. Notably, in high-bias environments where standard models collapse to negative R^2 (performing worse than predicting the mean), StablePFN maintains robust predictive power.

[1]Code and models are available at: https://github.com/GuanZhengkang/StablePFN

2 Related Work

2.1 Prior-Data Fitted Networks

PFNs redefine Bayesian posterior predictive distribution estimation as in-context learning, utilizing Transformers [21] trained on synthetic priors to approximate predictive distributions [27, 28]. TabPFN [17, 18] validates this paradigm on tabular data, outperforming tree-based baselines by leveraging priors derived from SCMs. Recent efforts scale this approach, TabPFN v2.5 [12] enhances context capacity via distillation, LimiX [50] enables joint distribution modeling, and Mitra [49] optimizes prior diversity.

Beyond architecture scaling, concurrent research scrutinizes the reliability of PFNs under distribution shifts and non-stationary settings. Ye et al. [46] analyze the internal mechanisms of TabPFN v2, revealing its capability as a discriminative feature extractor and extending it via test-time divide-and-conquer strategies. However, robustness remains a critical bottleneck. Evaluations in out-of-distribution (OOD) scenarios [6] and adversarial settings [9] highlight distinct vulnerabilities to domain shifts and test-time attacks. To address the limitations of the i.i.d. assumption [3], Helli et al. [16] introduce Drift-Resilient TabPFN, which adapts to temporal distribution shifts by incorporating time-variant SCM priors.

Additionally, this paradigm has demonstrated potential in causal effect estimation. Recent works such as CausalFM [25] and Do-PFN [33] show that PFNs can effectively model intervention outcomes. However, along with leading predictive models like TabPFN v2.5 [12] and LimiX [50], CausalFM and Do-PFN remain primarily focused on approximating (interventional) distributions through black-box fitting, lacking an explicit mechanism for structural causal awareness.

2.2 Causal and Markov Boundary Discovery

Causal discovery seeks to reconstruct causal graph structures from data. Local discovery focuses on the MB, the minimal set rendering a target independent of others [1], offering theoretically optimal feature selection [47]. While classical algorithms, such as Grow-Shrink (GS) [26] and Incremental Association Markov blanket (IA) [40] rely on conditional independence tests, modern variants improve efficiency (BAMB [23], EEMB [42]) and robustness (CCMB [43]), or address latent confounders [48].

For global structure learning, a shift from traditional methods (e.g., PC [38], LiNGAM [37]) occurred with the Dual-Axis Attention Block (DAB) in Non-Parametric Transformers [21]. By attending across both features and datapoints, DAB-based architectures underpin both PFNs and amortized causal discovery models like AVICI [24] and BCNP [8], which predict structures via supervised meta-learning. This synergy also extends to causal induction [20], active learning [2], and conditional independence testing [10].

2.3 Stable Prediction

Stable prediction addresses the challenge of prediction across unknown environments, thereby avoiding spurious correlations common in standard supervised learning [7]. Early methods [22, 35] combined prediction with global confounder balancing to ensure stable weights, while Shen et al. [36] employed sample reweighting to decorrelate inputs. Theoretical work later defined the *minimal*

stable variable set for optimal stable prediction [45]. Recently, the focus has shifted to explicit structural leverage. Wang et al. [41] use causal skeletons to separate invariant features, demonstrating that structural awareness is essential for robust generalization.

3 Preliminaries

3.1 PFN Formulation

We consider the problem of tabular prediction where the feature space is defined as $X \in \mathcal{X} \subset \mathbb{R}^F$ and the target variable as $Y \in \mathcal{Y} \subset \mathbb{R}$. An observed dataset $\mathcal{D}_n = \{(x_i, y_i)\}_{i=1}^n$ is sampled i.i.d. from a joint distribution $\mathbb{P}$. Within the non-parametric Bayesian framework, the joint distribution $\mathbb{P} \in \mathcal{P}$ is treated as a random probability measure sampled from a prior Π, denoted as $\mathbb{P} \sim \Pi$. Consequently, predicting the label y for a new query x given $\mathcal{D}_n$ is equivalent to computing the posterior predictive distribution (PPD):

$$p(y \mid x, \mathcal{D}_n) = \int_{\mathcal{P}} p(y \mid x, \mathbb{P}) d\Pi(\mathbb{P} \mid \mathcal{D}_n). \tag{1}$$

PFNs perform amortized inference on this PPD by parameterizing a neural network $p_\theta(y \mid x, \mathcal{D}_n)$ [28]. The optimal parameters are obtained by minimizing the expected negative log-likelihood (NLL) over the prior distribution:

$$\theta^* = \arg\min_\theta \mathbb{E}_{\mathbb{P}\sim\Pi} \mathbb{E}_{\mathcal{D}_n,(x,y)\sim\mathbb{P}}[-\log p_\theta(y \mid x, \mathcal{D}_n)], \tag{2}$$

where $\mathcal{D}_n$ and the test pair (x, y) are sampled independently from the same $\mathbb{P}$. In this framework, the prior Π is implicitly defined by the synthetic data generation mechanism used during pre-training. By constructing causal mechanisms that simulate complex real-world dependencies, PFNs learn to approximate the PPD for empirical data effectively.

3.2 Stable Prediction across Environments

Although PFNs incorporate rich causal structures during the synthetic pre-training phase, their default prediction mechanism utilizes the entire available feature set. In cross-environment scenarios, where the training dataset $\mathcal{D}_n$ and the test sample (x, y) are no longer identically distributed, spurious correlations between features and labels can lead to model failure.

To formalize this, we introduce the concept of *Environments*. Let $e \in \mathcal{E}$ denote a latent environmental variable, where each environment corresponds to a specific joint distribution $P^e(X, Y)$. While environments may vary, we posit that the underlying causal generation mechanism of the data remains invariant. Specifically:

Assumption 1 (Generative Mechanism Invariance)**.** *The underlying generative mechanism of the data is governed by an invariant Directed Acyclic Graph (DAG), where the functional relationships between variables remain invariant across all environments $e \in \mathcal{E}$.*

Under this assumption, the objective of **stable prediction** is defined as follows:

Problem 1 (Stable Prediction [22])**.** *Given a training dataset $\mathcal{D}^{tr}$ from a **single** environment $e_{tr} \in \mathcal{E}$, the goal is to learn a predictor p_θ that minimizes error fluctuation and maintains robust performance across any **unseen** testing environment $e_{te} \in \mathcal{E}$.*

Xu et al. [45] suggested that the essence of stable prediction lies in utilizing the minimal set of causal variables for prediction. This strategy effectively filters out spurious correlations arising from confounding or selection biases that shift across environments. Specifically, for modeling the conditional distribution of Y given X, a subset $S \subseteq X$ is considered the optimal stable predictor if and only if S constitutes the **MB** of Y in the training environment e_{tr} [45].

Definition 1 (Markov blanket & Markov boundary [31])**.** For a target variable Y and a feature set X, a subset $S \subseteq X$ is a Markov blanket of Y if $Y \perp\!\!\!\perp X \setminus S \mid S$. The MB is the unique minimal Markov blanket. In a faithful DAG, the MB of a node consists of its parents, children, and spouses (parents of its children).

Remark. It is important to note that if the objective were solely to minimize Mean Squared Error (MSE), the optimal predictor would rely on the *Minimal Stable Variable Set* [45], which is typically a subset of the MB. However, since the objective of PFNs is to model the full PPD rather than a simple point estimate, complete information regarding the conditional distribution is required. Therefore, we select the MB as the feature subset.

Consequently, within the PFN paradigm, we formulate stable prediction as a PPD estimation problem based on the MB. The model predicts using $p_\theta(y \mid x^{MB}, \mathcal{D}_n^{MB})$, where x^{MB} and $\mathcal{D}_n^{MB}$ represent the query and context data restricted to the MB features. In the following, we introduce StablePFN and detail its implementation for stable tabular prediction.

4 Method

4.1 Overall Framework

The architecture of StablePFN is built upon the DAB [21], a structure that has demonstrated its effectiveness in recent PFN-based tabular prediction [12, 18, 50] and causal discovery [8, 24] models. This structure captures both inter-feature dependencies and inter-sample relationships by alternating between Feature Attention and Sample Attention.

As illustrated in Figure 1, StablePFN adopts a multi-task shared-bottom architecture. It consists of a Shared Encoder with L_{sh} DAB layers, followed by three parallel functional branches: Decoupling, Discovery, and Prediction, containing L_{de}, L_{di}, and L_{pr} DAB layers, respectively. We consider a tabular regression task with training and testing sets for features and target variable: $X_{train} \in \mathbb{R}^{n_{tr}\times F}$, $y_{train} \in \mathbb{R}^{n_{tr}\times 1}$, $X_{test} \in \mathbb{R}^{n_{te}\times F}$, $y_{test} \in \mathbb{R}^{n_{te}\times 1}$, where $n_{tr} + n_{te} = N$. The inputs X_{train}, y_{train}, and X_{test} are first mapped into an embedding space of dimension E via linear projection. To maintain feature identity across different columns, random Feature ID Embeddings are added to the representations. The Shared Encoder then processes these embeddings to produce Global Representations $\mathbf{H} \in \mathbb{R}^{N\times(F+1)\times E}$, which are then routed to three parallel branches:

Decoupling Branch: This branch processes $\mathbf{H}_{de} \in \mathbb{R}^{n_{tr}\times F\times E}$, which corresponds to the representations of X_{train} within $\mathbf{H}$, to compute sample decoupling weights $\hat{w}$. Its primary objective is to reweight samples to minimize inter-feature dependencies. These weights are generated through an attention-based Aggregation Head.

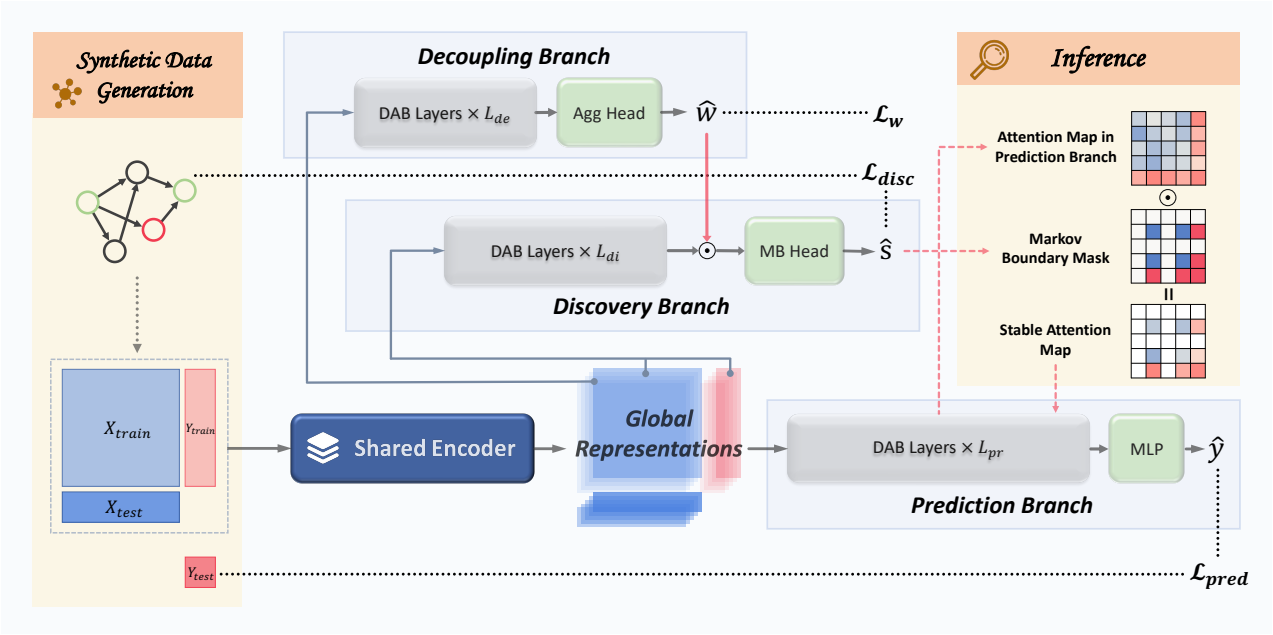


Figure 1: The overall architecture of StablePFN. The model adopts a multi-task shared-bottom framework comprising three branches: (1) The Decoupling Branch generates sample weights $\hat{w}$ to minimize feature dependencies; (2) The Discovery Branch identifies the Markov Boundary (MB) score $\hat{s}$, explicitly guided by the learned weights $\hat{w}$; (3) The Prediction Branch approximates the Posterior Predictive Distribution (PPD). During inference (right panel), the discovered structural knowledge $\hat{s}$ is converted into a Hard MB Mask, which is applied to the prediction branch's attention map (via element-wise product) to filter spurious correlations and ensure stable prediction.

Discovery Branch: This branch receives $\mathbf{H}_{di} \in \mathbb{R}^{n_{tr} \times (F+1) \times E}$, representing the combined embeddings of both X_{train} and y_{train} within $\mathbf{H}$. It aims to identify the MB of the target variable. Crucially, the discovery process is guided by the sample decoupling weights $\hat{w}$ within a specialized Weighted MB Head.

Prediction Branch: This branch utilizes the full Global Representations $\mathbf{H}$ for ICL to model the PPD. During training, it is optimized to approximate the PPD of y_{te} using the full feature set. During inference, it incorporates the MB Mask generated by the Discovery Branch to facilitate stable prediction.

4.1.1 Feature-to-Weight Aggregation Head. The primary objective of the Decoupling Branch is to derive a set of sample weights $\hat{w}$ based on the feature distribution, ensuring that features become as independent as possible after reweighting. To achieve this, we introduce an Attention Aggregator based on cross-attention in the Aggregation Head.

Instead of employing simple average pooling, this module utilizes a query token $\mathbf{q} \in \mathbb{R}^E$ to adaptively retrieve information across the feature dimension. Given the representations $\mathbf{H}^{out} \in \mathbb{R}^{n_{tr} \times F \times E}$ output by the DAB layers of the Decoupling Branch, the compressed representation $\mathbf{z}_i$ for each sample i is computed as follows:

$$\mathbf{z}_i = \text{LayerNorm}(\text{MHA}(\mathbf{q}, \mathbf{H}^{out}_{i,\cdot,\cdot}, \mathbf{H}^{out}_{i,\cdot,\cdot})), \tag{3}$$

where MHA denotes Multi-Head Attention, with $\mathbf{q}$ acting as the query and the feature embeddings $\mathbf{H}^{out}_{i,\cdot,\cdot} \in \mathbb{R}^{F \times E}$ serving as both keys and values. This mechanism enables the model to adaptively aggregate feature-wise information most critical for sample decoupling. Finally, $\mathbf{z}_i$ is passed through an MLP layer to generate the final sample weight $\hat{w}_i$.

4.1.2 Weighted MB Head. The Discovery Branch aims to identify the MB of the target variable. Within the Weighted MB Head, we utilize the estimated decoupling weights $\hat{w}$ to guide the computation of associations between features and the target.

To prevent gradients from the Discovery Branch from interfering with the optimization of the Decoupling Branch, we treat the weights as constants by applying a stop-gradient operator $\text{sg}[\cdot]$. We then compute the weighted similarity between the feature and target embeddings. Let $\mathbf{x}_j \in \mathbb{R}^{F \times E}$ and $\mathbf{y}_j \in \mathbb{R}^{1 \times E}$ denote the representations for sample j within the output of the DAB layers of the Discovery Branch. The interaction terms are computed as follows:

$$\begin{aligned} \mathrm{S}_{xy} &= \sum_{j=1}^{n_{tr}} \text{sg}[\hat{w}_j] \cdot (\mathbf{x}_j \odot \mathbf{y}_j), \\ \mathrm{S}_{xx} &= \sum_{j=1}^{n_{tr}} \text{sg}[\hat{w}_j] \cdot (\mathbf{x}_j \odot \mathbf{x}_j), \\ \mathrm{S}_{yy} &= \sum_{j=1}^{n_{tr}} \text{sg}[\hat{w}_j] \cdot (\mathbf{y}_j \odot \mathbf{y}_j), \end{aligned} \tag{4}$$

where $\odot$ denotes the element-wise (Hadamard) product with broadcasting over the feature dimension F. Here, S_{xy} captures the weighted cross-correlation in the embedding space, while S_{xx} and

$\mathbf{S}_{yy}$ serve as normalization metrics that account for feature and target intensities. Then we apply a signed logarithmic transformation to these metrics:

$$\mathbf{F}_{comb} = \text{Concat}\big(\text{sign}(\mathbf{S}_{xy})\log(1+|\mathbf{S}_{xy}|), \log(1+\mathbf{S}_{xx}),\ \text{Tile}(\log(1+\mathbf{S}_{yy}))\big), \tag{5}$$

where $\text{Tile}(\cdot)$ replicates the target intensity metrics across the F dimension to ensure compatibility. The resulting vector $\mathbf{F}_{comb} \in \mathbb{R}^{F\times 3E}$ is subsequently processed by an MLP with a Sigmoid activation to produce the probability score $\hat{\mathbf{s}} \in [0,1]^F$, representing the likelihood of each feature belonging to the MB.

4.2 Multi-task Pre-training

4.2.1 Prior. Our prior data generation follows the methodology of SCMs, aligning with previous works [12, 18, 50]. For each synthetic dataset, we first sample a random number of nodes and construct a DAG using the Erdős-Rényi model. The structural equations for each node are then formed by stochastically selecting from a diverse set of functional components, including random neural networks with varied activation functions, decision tree structures, and discretization steps. Gaussian noise with randomized variance is subsequently added to the system. To simulate real-world observational noise, we apply distortion and binning to the observed variables. Finally, a node is randomly designated as the target variable y.

Unlike previous works that focus exclusively on end-to-end prediction, we leverage the underlying ground-truth DAG during generation to derive the true MB labels for the target variable y in each synthetic dataset. This provides the necessary supervision for our multi-task learning framework.

4.2.2 Training Objectives and Loss Functions.

Decoupling Loss. The objective of the decoupling branch is to learn a global sample weight vector $\hat{w} \in \mathbb{R}^{n_{tr}}$ that renders features mutually independent. Rather than global decorrelation, we design a **Weighted Local Covariance Loss** computed via the following steps:

1. Local Sampling: Given a total of T sampling iterations, in each iteration t, we randomly select a hyper-rectangular subspace within the feature space. Let $\mathcal{I}_t$ denote the set of indices for the samples falling within this subspace.

2. Weighted Correlation: For the t-th subset, we utilize the corresponding weights $\{\hat{w}_i \mid i \in \mathcal{I}_t\}$ to compute the weighted correlation matrix $\mathbf{R}^{(t)}$. The element (j,k) of this matrix is defined as:

$$\mathbf{R}^{(t)}_{jk} = \frac{\text{Cov}_t(x_j, x_k)}{\sqrt{\text{Var}_t(x_j)\cdot \text{Var}_t(x_k)}}, \tag{6}$$

where Cov_t and Var_t denote the weighted covariance and variance computed exclusively over the samples in $\mathcal{I}_t$.

3. Regularization: To prevent weight degeneracy (i.e., weights concentrating on a few samples), we introduce a squared sum regularization term on the global weights.

The final decoupling loss $\mathcal{L}_w$ is defined as the average of the squared off-diagonal elements of the correlation matrices across all subspaces, combined with the regularization term:

$$\mathcal{L}_w = \frac{1}{T}\sum_{t=1}^{T}\left(\frac{1}{F(F-1)}\sum_{j\neq k}(\mathbf{R}^{(t)}_{jk})^2\right) + \lambda_{\text{reg}}\|\hat{w}\|^2. \tag{7}$$

Discovery Loss. For the discovery branch, given the probability score $\hat{\mathbf{s}} \in [0,1]^F$ output by the Weighted MB Head, we employ the standard Binary Cross-Entropy (BCE) loss:

$$\mathcal{L}_{disc} = -\frac{1}{F}\sum_{f=1}^{F}[g_f \log(s_f) + (1-g_f)\log(1-s_f)], \tag{8}$$

where s_f is the f-th element of $\hat{\mathbf{s}}$, and $g_f \in \{0,1\}$ is the ground-truth MB label derived from the synthetic graph structure.

Predictive Loss. We approximate the PPD $p_\theta(y \mid x, \mathcal{D}_n)$ using a categorical distribution over discretized bins [18]. We discretize the continuous target space $\mathcal{Y}$ into B intervals defined by boundaries $\mathbf{b} = \{b_0, \ldots, b_B\}$. For a given target y, let k denote the index of the interval such that $y \in [b_k, b_{k+1})$. The probability density is modeled as a piece-wise constant function derived from the model's logit output $\mathbf{z} \in \mathbb{R}^B$:

$$\log p_\theta(y \mid x, \mathcal{D}_n) = \log\left(\frac{\exp(z_k)}{\sum_j \exp(z_j)}\right) - \log(b_{k+1} - b_k). \tag{9}$$

To ensure robustness against OOD targets falling outside $[b_0, b_B]$, we incorporate a log-space linear penalty term proportional to the distance from the nearest boundary. The final predictive loss minimizes the NLL over the synthetic query set $\mathcal{D}_{test}$:

$$\mathcal{L}_{pred} = -\mathbb{E}_{(x,y)\sim\mathcal{D}_{test}}\left[\log p_{\text{adj}}(y \mid x, \mathcal{D}_n)\right]. \tag{10}$$

The final joint loss for our end-to-end multi-task training is:

$$\mathcal{L}_{total} = \lambda_1\mathcal{L}_w + \lambda_2\mathcal{L}_{disc} + \lambda_3\mathcal{L}_{pred}. \tag{11}$$

4.3 Stable Inference via MB Attention Constraints

While the training phase jointly optimizes the sample decoupling weights $\hat{w}$ and the MB scores $\hat{\mathbf{s}}$, the ultimate objective is to achieve stable prediction during inference. In this section, we describe how to explicitly inject the learned structural knowledge into the Prediction Branch to approximate the stable posterior $p_\theta(y \mid x^{MB}, \mathcal{D}_n^{MB})$.

As illustrated in the right panel of Figure 1, a standard DAB's feature-wise attention computes global attention across the entire feature set. Inspired by recent advances in causal attention mechanisms [14, 15, 29, 34, 44], StablePFN employs a hard MB masking mechanism during the inference stage to filter out spurious correlations.

Specifically, we derive a binary MB Mask $\mathbf{m} = \mathbb{I}(\hat{\mathbf{s}} > 0.5) \in \{0,1\}^F$ based on the probability scores $\hat{\mathbf{s}}$ output by the Discovery Branch (a detailed sensitivity analysis regarding this threshold is provided in Appendix B.1). To accommodate the target variable in the attention map, we extend this mask to $\mathbf{M}_{MB} = [\mathbf{m}; 1] \in \{0,1\}^{F+1}$. This mask is then applied to the feature attention matrix of the Prediction Branch as a structural constraint. Let $\mathbf{A}_{pred} \in \mathbb{R}^{N\times(F+1)\times(F+1)}$ denote the feature attention map of DAB in the Prediction Branch. The stabilized attention is computed via an element-wise Hadamard product:

$$\mathbf{A}_{stable} = \mathbf{A}_{pred} \odot \left(\mathbf{M}_{MB}\mathbf{M}_{MB}^\top\right). \tag{12}$$

We apply this MB Mask to the feature attention matrices of all DAB layers within the Prediction Branch. This effectively constrains the model to attend solely to the MB features and the target variable, thereby achieving stable prediction during inference.

5 Experiments

In this section, we present comprehensive results to demonstrate the effectiveness of StablePFN in stable prediction. We emulate spurious correlations that typically arise during real-world data collection on both synthetic and real-world datasets, evaluating StablePFN's performance in these challenging environments.

5.1 Experimental Setup

Following the established experimental design in stable learning [22, 36], we construct training and testing environments with distinct statistical properties by subjecting the original datasets to feature injection followed by biased sampling. This setup simulates the spurious correlations caused by potential selection biases in real-world scenarios. Specifically, we process an original dataset $\mathcal{D}_{real} = \{\mathbf{X}_{real}, \mathbf{y}_{real}\}$ as follows:

Spurious Feature Injection. First, we introduce controlled spurious correlations into the original features. Given a sample size N, we generate a latent variable vector $\mathbf{z} \in \mathbb{R}^N$, where each element is drawn from a standard normal distribution, i.e., $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_N)$. We then construct the observed spurious features $\mathbf{X}_s$. Assuming we select a subset of dimensions to be spurious, we define:

$$\mathbf{X}_s = \mathbf{z}\mathbf{1}^\top + \epsilon, \tag{13}$$

where $\mathbf{1}$ is a vector of ones matching the dimension of the selected features, and ϵ represents noise sampled from $\mathcal{N}(0, \sigma^2)$ with $\sigma^2 = 0.1$. Let F denote the total number of feature dimensions of $\mathbf{X}_{real}$. We control the proportion of spurious features using the ratio $\rho \in [0, 1]$. Specifically, we randomly select $K = \lceil F \cdot \rho \rceil$ dimensions from the original features $\mathbf{X}_{real}$ and replace them with the columns of $\mathbf{X}_s$ to obtain the combined feature set $\mathbf{X}_{combined}$.

Biased Sampling. To simulate environmental discrepancies, we employ an exponential weighted sampling strategy based on the Euclidean distance between y and z, analogous to a Gaussian kernel. For the training environment e_{tr}, we induce a positive correlation between the latent variable $\mathbf{z}$ and the target variable y. The sampling probability for the training set is defined as:

$$P(i \in e_{tr}) \propto \exp(-\|\hat{y}_i - \hat{z}_i\|^2), \tag{14}$$

where $\hat{y}_i$ and $\hat{z}_i$ are the standardized samples of the target and latent variables, respectively. Conversely, for the testing environment e_{te}, we invert this spurious correlation. The sampling probability for the test set is defined as:

$$P(j \in e_{te}) \propto \exp(-\|\hat{y}_j + \hat{z}_j\|^2). \tag{15}$$

Since $\mathbf{X}_s$ is derived from $\mathbf{z}$ with additive noise, the induced statistical dependency between $\mathbf{y}_{real}$ and $\mathbf{z}$ directly propagates to $\mathbf{X}_s$. In e_{tr}, $\mathbf{X}_s$ exhibits a strong positive correlation with the label, creating a misleading "shortcut" signal that conventional models tend to overfit. In e_{te}, this correlation is inverted. Consequently, models relying on spurious features will suffer significant performance degradation. This severe distribution shift compels the model to discard environment-sensitive spurious features and instead rely on the invariant causal mechanisms within $\mathbf{X}_{real}$.

Baselines and Metrics. For the regression-based stable prediction task, we compare StablePFN against three categories of baselines: (1) an ablation variant, StablePFN (Unmasked), to evaluate the necessity of the structural constraint; (2) current state-of-the-art PFN-based foundation models, including **TabPFN v2.5** [12] and **LimiX (16M)** [50]; and (3) widely-used traditional methods such as LightGBM [19] and LASSO [39]. To ensure a fair comparison of the intrinsic model capabilities, we disable all ensemble mechanisms and evaluate all PFN-based models in a single-pass inference mode. We report the Mean Squared Error (MSE) and the Coefficient of Determination ($R^2 \in (-\infty, 1]$) as evaluation metrics. While $R^2 = 1$ indicates perfect prediction and $R^2 = 0$ indicates a baseline prediction of the mean, R^2 may become negative, as it indicates that the model's predictions are worse than simply guessing the target mean, revealing that the model has overfitted to unstable spurious correlations and failed to generalize to the new environment. To account for the stochasticity in our spurious correlation sampling, all reported results are averaged over 100 independent trials.

Table 1: Performance (AUC) comparison of Markov Boundary (MB) Discovery on synthetic datasets. For global causal discovery baselines (PC, LiNGAM, AVICI, BCNP), we first reconstruct the complete causal graph and then extract the MB of the target variable for evaluation against the ground truth. Best results are bolded.

Method	10 Nodes		20 Nodes	
	$N = 800$	$N = 2000$	$N = 800$	$N = 2000$
PC	0.556	0.555	0.562	0.570
LiNGAM	0.546	0.550	0.572	0.565
GS	0.592	0.591	0.598	0.598
IA	0.610	0.612	0.627	0.631
AVICI	0.528	0.538	0.536	0.506
BCNP	0.516	0.508	0.533	0.531
StablePFN	**0.627**	**0.632**	**0.630**	**0.638**

5.2 Synthetic Datasets

To demonstrate StablePFN's ability to discover MB, we first conduct experiments on synthetic datasets where the ground-truth causal graph is known. To ensure fairness and demonstrate generalization, the generation mechanism for these datasets differs from StablePFN's pre-training prior. We generate random DAGs but define the relationships between parent and child nodes using random non-linear functions from the set $\{x, |x|, x^2, x^3, \tanh(x), \sin(x)\}$, superimposed with Gaussian noise of random magnitudes. We conduct experiments across four settings, combining node counts of 10 and 20, and sample sizes of $N = 800$ and $N = 2000$.

5.2.1 MB Discovery on Synthetic Datasets. For the MB discovery task, we compare our method against two categories of baselines: (1) Traditional Causal and MB Discovery algorithms, encompassing global discovery methods PC [38] (constraint-based) and LiNGAM [37] (functional-based), alongside local MB discovery algorithms GS [26] and IA [40]; (2) Amortized Causal Discovery methods, including AVICI [24] and BCNP [8]. To ensure a rigorous comparison, we adhere to the standard evaluation protocols for AVICI and BCNP, training them specifically on the data-generating mechanisms of our synthetic datasets to maximize their performance. In contrast, StablePFN is evaluated in a strictly zero-shot

Table 2: Performance comparison for stable prediction on synthetic datasets (R^2 and MSE, Mean ± Std), best results are bolded.

Setting	Sampling Bias	Metric	StablePFN	StablePFN (Unmasked)	TabPFN v2.5	LimiX	LightGBM	LASSO
10 Nodes, N=800	Moderate ($k=2$)	R^2	**$0.614_{\pm 0.40}$**	$0.423_{\pm 0.64}$	$0.508_{\pm 0.62}$	$0.497_{\pm 0.60}$	$0.355_{\pm 0.67}$	$0.149_{\pm 0.69}$
		MSE	**$0.305_{\pm 0.37}$**	$0.453_{\pm 0.59}$	$0.385_{\pm 0.57}$	$0.392_{\pm 0.56}$	$0.493_{\pm 0.61}$	$0.627_{\pm 0.60}$
	High ($k=10$)	R^2	**$0.516_{\pm 0.48}$**	$0.271_{\pm 0.67}$	$0.385_{\pm 0.64}$	$0.375_{\pm 0.62}$	$0.209_{\pm 0.71}$	$0.059_{\pm 0.70}$
		MSE	**$0.278_{\pm 0.30}$**	$0.420_{\pm 0.43}$	$0.354_{\pm 0.41}$	$0.361_{\pm 0.40}$	$0.450_{\pm 0.46}$	$0.540_{\pm 0.46}$
10 Nodes, N=2000	Moderate ($k=2$)	R^2	**$0.675_{\pm 0.35}$**	$0.487_{\pm 0.56}$	$0.504_{\pm 0.58}$	$0.511_{\pm 0.55}$	$0.396_{\pm 0.64}$	$0.230_{\pm 0.68}$
		MSE	**$0.262_{\pm 0.32}$**	$0.426_{\pm 0.51}$	$0.414_{\pm 0.53}$	$0.407_{\pm 0.51}$	$0.500_{\pm 0.59}$	$0.620_{\pm 0.62}$
	High ($k=10$)	R^2	**$0.637_{\pm 0.36}$**	$0.434_{\pm 0.62}$	$0.480_{\pm 0.64}$	$0.470_{\pm 0.63}$	$0.351_{\pm 0.68}$	$0.114_{\pm 0.71}$
		MSE	**$0.217_{\pm 0.26}$**	$0.345_{\pm 0.44}$	$0.315_{\pm 0.44}$	$0.322_{\pm 0.44}$	$0.395_{\pm 0.48}$	$0.529_{\pm 0.51}$
20 Nodes, N=800	Moderate ($k=2$)	R^2	$0.391_{\pm 0.56}$	$0.281_{\pm 0.67}$	**$0.423_{\pm 0.65}$**	$0.402_{\pm 0.62}$	$0.239_{\pm 0.71}$	$0.135_{\pm 0.70}$
		MSE	$0.472_{\pm 0.49}$	$0.547_{\pm 0.59}$	**$0.435_{\pm 0.56}$**	$0.451_{\pm 0.55}$	$0.570_{\pm 0.61}$	$0.666_{\pm 0.63}$
	High ($k=10$)	R^2	**$0.354_{\pm 0.65}$**	$0.126_{\pm 0.84}$	$0.302_{\pm 0.75}$	$0.282_{\pm 0.73}$	$0.106_{\pm 0.85}$	$-0.017_{\pm 0.76}$
		MSE	**$0.373_{\pm 0.44}$**	$0.491_{\pm 0.53}$	$0.391_{\pm 0.48}$	$0.400_{\pm 0.46}$	$0.499_{\pm 0.54}$	$0.561_{\pm 0.49}$
20 Nodes, N=2000	Moderate ($k=2$)	R^2	**$0.520_{\pm 0.45}$**	$0.380_{\pm 0.60}$	$0.488_{\pm 0.59}$	$0.461_{\pm 0.57}$	$0.335_{\pm 0.66}$	$0.091_{\pm 0.72}$
		MSE	**$0.394_{\pm 0.42}$**	$0.496_{\pm 0.54}$	$0.415_{\pm 0.54}$	$0.436_{\pm 0.52}$	$0.532_{\pm 0.60}$	$0.729_{\pm 0.67}$
	High ($k=10$)	R^2	**$0.570_{\pm 0.45}$**	$0.358_{\pm 0.68}$	$0.459_{\pm 0.65}$	$0.434_{\pm 0.64}$	$0.309_{\pm 0.72}$	$0.153_{\pm 0.75}$
		MSE	**$0.254_{\pm 0.29}$**	$0.372_{\pm 0.44}$	$0.316_{\pm 0.42}$	$0.327_{\pm 0.42}$	$0.400_{\pm 0.47}$	$0.481_{\pm 0.49}$

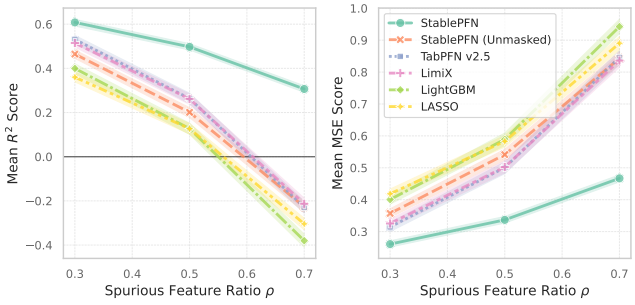


Figure 2: Performance comparison under varying spurious feature ratio ρ. The lines represent the mean R^2 and mean MSE across all settings, with the 95% confidence intervals (shaded areas).

manner, highlighting the superior generalization capability of our model. Table 1 summarizes the MB discovery results in terms of AUC, while the comprehensive evaluation metrics are provided in Appendix C. StablePFN consistently achieves the highest AUC scores across all settings, and demonstrates greater scalability and robustness compared to existing end-to-end amortized baselines. For instance, in the 20-node setting ($N = 2000$), while amortized baselines AVICI and BCNP struggle to maintain performance, with AUCs hovering between 0.50 and 0.53, StablePFN maintains a high AUC of 0.638.

5.2.2 Stable Prediction on Synthetic Datasets. We evaluate performance across four settings, varying the sampling bias intensity $k \in \{2, 10\}$ to control the strength of the induced spurious associations. Here, k represents the multiplier for the candidate pool size relative to the target sample size during the biased sampling process. Consequently, a larger k (denoted as *High* in Table 2) implies a larger pool from which to select biased samples, creating a sharper divergence between training and testing environments. The results are detailed in Table 2.

Performance and Robustness. StablePFN achieves the best metrics in 7 out of 8 settings. While traditional baselines (LightGBM, LASSO) collapse to near-zero or negative R^2 in high bias scenarios ($k = 10$) due to overfitting spurious correlations, StablePFN exhibits remarkable stability. For instance, in the challenging 20-node setting with high bias ($k = 10$), it outperforms the strongest baseline, TabPFN v2.5 (0.354 vs 0.302 in R^2). This confirms that by integrating causal variable selection, StablePFN effectively bridges the gap between local causal discovery and predictive modeling.

Impact of MB Mask. The ablation study validates the necessity of our explicit structural constraint. Removing the MB Mask leads to consistent performance degradation, particularly in high bias settings (e.g., 10 Nodes, $k = 10$), where R^2 drops significantly from 0.516 to 0.271. This result demonstrates that unconstrained end-to-end PPD estimation alone is insufficient for cross-environment generalization. It is the explicit restriction of the attention mechanism to the MB that filters spurious pathways.

Impact of Spurious Feature Ratio ρ. Furthermore, we analyze the impact of the spurious feature ratio ρ on model performance. As shown in Figure 2, the difficulty of the prediction task increases monotonically with ρ. When the ratio of spurious features is low ($\rho = 0.3$), most models achieve reasonable performance, with StablePFN holding a slight lead. However, as the environment becomes increasingly dominated by spurious correlations ($\rho \to 0.7$), the performance of baseline models, including the unmasked variant of StablePFN, degrades rapidly.

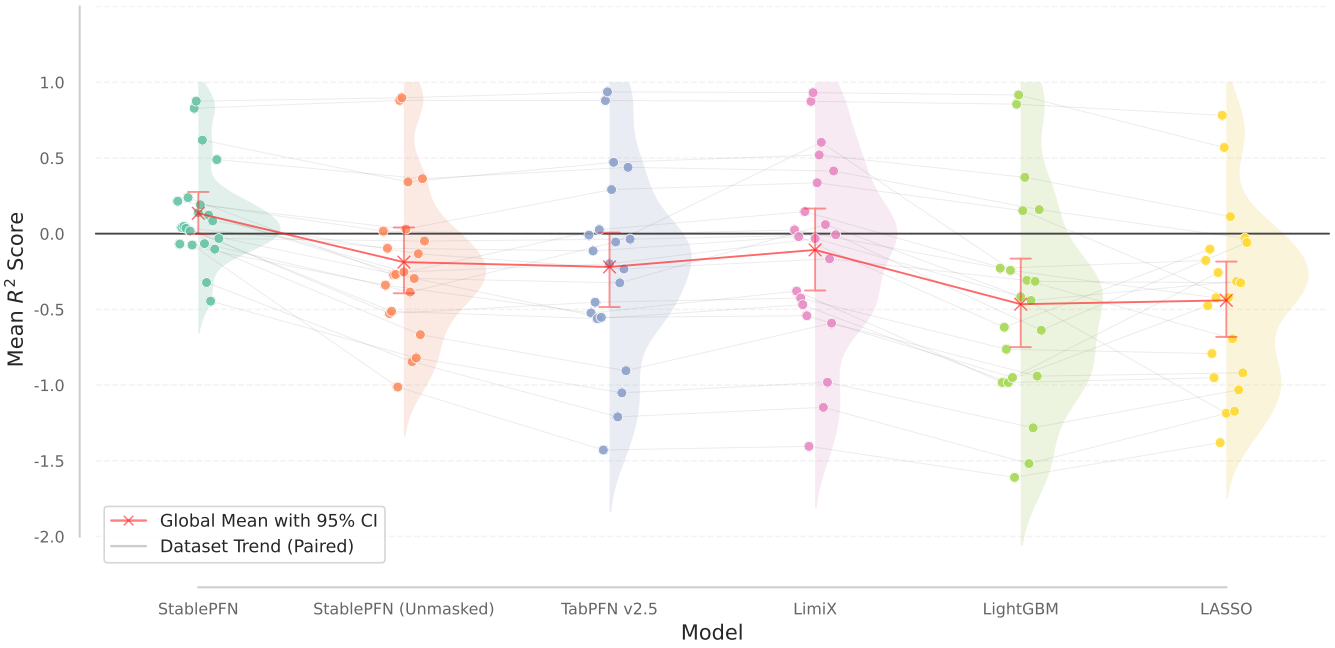


Figure 3: Performance distribution on 21 real-world datasets under spurious feature injection ($\rho = 0.3$). The violin plots illustrate the overall density, while scattered dots represent results for individual datasets. Grey lines track the paired performance changes across models. Red markers and error bars indicate the global mean and 95% confidence intervals.

At the extreme setting of $\rho = 0.7$, traditional methods and standard PFNs collapse to negative R^2 scores, indicating predictive failure that is even worse than mean prediction. In sharp contrast, StablePFN maintains a much flatter degradation curve. Even when 70% of the input features are spurious, StablePFN retains valid predictive capability ($R^2 \approx 0.3$), demonstrating that its MB discovery mechanism successfully identifies and relies on the sparse invariant causal signals amidst overwhelming noise.

5.3 Real-world Datasets

Dataset Collection. To ensure an objective evaluation, we curated a benchmark by systematically filtering the OpenML repository based on three pre-defined, exhaustive criteria: (1) feature count $F \in [5, 20]$; (2) sample size $N < 20,000$; and (3) exclusion of all synthetic (e.g., "simulated", "BNG") or duplicate entries. This process yielded a collection of 21 high-quality real-world datasets.

Experimental Setup. We applied the same spurious feature injection and biased sampling protocol described in Section 5.1 to these datasets. We evaluated model performance under spurious feature ratios of $\rho \in \{0.3, 0.5, 0.7\}$. Due to space constraints, we present the results for $\rho = 0.3$ in the main text, while the visualizations for $\rho = 0.5$ and $\rho = 0.7$, along with the complete tabular results for R^2 and MSE are provided in the Appendix C.

Generalization Capability. As shown in Figure 3, real-world data presents a significantly harder challenge than synthetic benchmarks. While standard models collapse, StablePFN demonstrates superior robustness, being the only method to maintain a positive global mean R^2 (≈ 0.2) with a tight confidence interval. Crucially, StablePFN avoids the catastrophic "performance crash" seen in other models. This shows that our model successfully distinguishes true causal signals from misleading environmental noise.

Baseline Failure. The results reveal two distinct failure modes. First, Traditional algorithms (LightGBM, LASSO) suffer from severe overfitting to spurious correlations, resulting in deep negative R^2 scores. This confirms that without explicit causal regularization, these models treat the injected spurious features, which are strongly correlated in training but reversed in testing, as valid predictive signals, leading to "anti-learning" in the target environment. Second, state-of-the-art PFN-based models (TabPFN v2.5, LimiX) also show limited cross-environment generalization, hovering near zero R^2. This suggests that large-scale pre-training alone, without explicit structural constraints like the MB mask, is insufficient for maintaining stable performance across varying environments.

Ablation Validation. The performance gap between StablePFN and StablePFN (Unmasked) is statistically significant. The unmasked version follows the trend of other baselines, dropping to near-zero performance. This validates our hypothesis: implicit causal learning alone may not guarantee the effective utilization of causal information under varying environments. The explicit structural constraint provided by the MB mask is critical for filtering out environment-sensitive spurious correlations.

Furthermore, to isolate the specific contribution of the sample reweighting mechanism, we present an additional ablation study focusing on the Decoupling Branch in Appendix B.2, which confirms its foundational role in our framework.

Computational Cost. Finally, to address the practical efficiency of our multi-branch architecture, we evaluate the computational cost and time complexity in Appendix B.3. Empirical results demonstrate that StablePFN maintains highly competitive memory footprints and inference speeds compared to standard baseline models.

6 Discussion

In this work, we introduced StablePFN, a causal-aware tabular foundation model that bridges the gap between powerful distribution fitting and mechanistic causal understanding. By employing a "Decouple-Discover-Predict" architecture, StablePFN explicitly identifies the MB of target variables and leverages this information to achieve stable prediction through hard MB masking. Evaluations on synthetic and 21 real-world datasets under spurious feature injection show that StablePFN significantly outperforms state-of-the-art models like TabPFN v2.5 and LimiX. While standard models collapse to negative R^2 in challenging scenarios, StablePFN maintains robust predictive power. To further advance this paradigm, several limitations invite future exploration:

Causal Framework and Mechanism Invariance: It is important to emphasize that StablePFN is fundamentally grounded in the structural causal framework [31]. All of our analyses operate under the premise that there exists a well-defined, underlying causal structure (representable as a DAG) that governs the data-generating process. Building upon this, we further assume that the underlying functional relationships on the DAG remain invariant across environments. In highly non-stationary real-world systems, the causal mechanisms themselves might shift, or such a definitive structure might not even exist. Extending StablePFN to detect and adapt to mechanism shifts is a significant next step.

Error Propagation from (Local) Causal Discovery: Upstream MB discovery is not infallible. If the Discovery Branch yields false positives (including spurious features) or false negatives (missing true causes), the subsequent prediction may fail to filter spurious shortcuts or may discard critical causal information. While this "cascading error" problem has been discussed in causal effect estimation [4, 11, 30], its impact on high-dimensional tabular foundation models remains to be fully quantified. Future work could investigate uncertainty-aware attention to buffer against structural misidentification.

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References

[1] Constantin F. Aliferis, Alexander Statnikov, Ioannis Tsamardinos, Subramani Mani, and Xenofon D. Koutsoukos. 2010. Local Causal and Markov Blanket Induction for Causal Discovery and Feature Selection for Classification Part I: Algorithms and Empirical Evaluation. *Journal of Machine Learning Research* 11, 7 (2010), 171–234. http://jmlr.org/papers/v11/aliferis10a.html

[2] Yashas Annadani, Panagiotis Tigas, Stefan Bauer, and Adam Foster. 2024. Amortized Active Causal Induction with Deep Reinforcement Learning. In *ICML 2024 Workshop on Structured Probabilistic Inference & Generative Modeling*. https://openreview.net/forum?id=81w8cEGkDA

[3] Haorun Cai and Han-Jia Ye. 2025. Understanding the Limits of Deep Tabular Methods with Temporal Shift. In *Proceedings of the 42nd International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 267)*, Aarti Singh, Maryam Fazel, Daniel Hsu, Simon Lacoste-Julien, Felix Berkenkamp, Tegan Maharaj, Kiri Wagstaff, and Jerry Zhu (Eds.). PMLR, 6366–6386. https://proceedings.mlr.press/v267/cai25j.html

[4] T Chang, Z Guo, and D Malinsky. 2025. Post-selection inference for causal effects after causal discovery. *Biometrika* (10 2025), asaf073. doi:10.1093/biomet/asaf073

[5] Tianqi Chen and Carlos Guestrin. 2016. XGBoost: A Scalable Tree Boosting System. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (San Francisco, California, USA) *(KDD '16)*. Association for Computing Machinery, New York, NY, USA, 785–794. doi:10.1145/2939672.2939785

[6] Zi-Jian Cheng, Zi-Yi Jia, Zhi Zhou, Yu-Feng Li, and Lan-Zhe Guo. 2025. Realistic Evaluation of TabPFN v2 in Open Environments. arXiv:2505.16226 [cs.LG] https://arxiv.org/abs/2505.16226

[7] Peng Cui and Susan Athey. 2022. Stable learning establishes some common ground between causal inference and machine learning. *Nature Machine Intelligence* 4, 2 (Feb. 2022), 110–115. doi:10.1038/s42256-022-00445-z

[8] Anish Dhir, Matthew Ashman, James Requeima, and Mark van der Wilk. 2025. A Meta-Learning Approach to Bayesian Causal Discovery. In *The Thirteenth International Conference on Learning Representations*. https://openreview.net/forum?id=eeJz7eDWKO

[9] Mohamed Djilani, Thibault Simonetto, Karim Tit, Florian Tambon, Paul Récamier, Salah Ghamizi, Maxime Cordy, and Mike Papadakis. 2025. On the Robustness of Tabular Foundation Models: Test-Time Attacks and In-Context Defenses. arXiv:2506.02978 [cs.LG] https://arxiv.org/abs/2506.02978

[10] Bao Duong, Nu Hoang, and Thin Nguyen. 2025. Amortized Conditional Independence Testing. In *Advances in Knowledge Discovery and Data Mining*, Xintao Wu, Myra Spiliopoulou, Can Wang, Vipin Kumar, Longbing Cao, Yanqiu Wu, Yu Yao, and Zhangkai Wu (Eds.). Springer Nature Singapore, Singapore, 410–423.

[11] Paula Gradu, Tijana Zrnic, Yixin Wang, and Michael I. Jordan. 2025. Valid Inference After Causal Discovery. *J. Amer. Statist. Assoc.* 120, 550 (2025), 1127–1138. arXiv:https://doi.org/10.1080/01621459.2024.2402089 doi:10.1080/01621459.2024.2402089

[12] Léo Grinsztajn, Klemens Flöge, Oscar Key, Felix Birkel, Philipp Jund, Brendan Roof, Benjamin Jäger, Dominik Safaric, Simone Alessi, Adrian Hayler, Mihir Manium, Rosen Yu, Felix Jablonski, Shi Bin Hoo, Anurag Garg, Jake Robertson, Magnus Bühler, Vladyslav Moroshan, Lennart Purucker, Clara Cornu, Lilly Charlotte Wehrhahn, Alessandro Bonetto, Bernhard Schölkopf, Sauraj Gambhir, Noah Hollmann, and Frank Hutter. 2025. TabPFN-2.5: Advancing the State of the Art in Tabular Foundation Models. arXiv:2511.08667 [cs.LG] https://arxiv.org/abs/2511.08667

[13] Léo Grinsztajn, Edouard Oyallon, and Gaël Varoquaux. 2022. Why do tree-based models still outperform deep learning on typical tabular data?. In *Proceedings of the 36th International Conference on Neural Information Processing Systems* (New Orleans, LA, USA) *(NIPS '22)*. Curran Associates Inc., Red Hook, NY, USA, Article 37, 14 pages.

[14] Kairong Han, Nuanqiao Shan, Ziyu Zhao, Zijing Hu, Xinpeng Dong, Junjian Ye, Lujia Pan, Fei Wu, and Kun Kuang. 2025. C^2DLM: Causal Concept-Guided Diffusion Large Language Models. arXiv:2511.22146 [cs.CL] https://arxiv.org/abs/2511.22146

[15] Kairong Han, Wenshuo Zhao, Ziyu Zhao, Ye Jun Jian, Lujia Pan, and Kun Kuang. 2025. CAT: Causal Attention Tuning For Injecting Fine-grained Causal Knowledge into Large Language Models. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, Christos Christodoulopoulos, Tanmoy Chakraborty, Carolyn Rose, and Violet Peng (Eds.). Association for Computational Linguistics, Suzhou, China, 9904–9921. doi:10.18653/v1/2025.emnlp-main.502

[16] Kai Helli, David Schnurr, Noah Hollmann, Samuel Müller, and Frank Hutter. 2024. Drift-Resilient TabPFN: In-Context Learning Temporal Distribution Shifts on Tabular Data. In *Advances in Neural Information Processing Systems*, A. Globerson, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. Tomczak, and C. Zhang (Eds.), Vol. 37. Curran Associates, Inc., 98742–98781. doi:10.52202/079017-3134

[17] Noah Hollmann, Samuel Müller, Katharina Eggensperger, and Frank Hutter. 2023. TabPFN: A Transformer That Solves Small Tabular Classification Problems in a Second. In *The Eleventh International Conference on Learning Representations*. https://openreview.net/forum?id=cp5PvcI6w8_

[18] Noah Hollmann, Samuel Müller, Lennart Purucker, Arjun Krishnakumar, Max Körfer, Shi Bin Hoo, Robin Tibor Schirrmeister, and Frank Hutter. 2025. Accurate predictions on small data with a tabular foundation model. *Nature* 637, 8045 (Jan. 2025), 319–326. doi:10.1038/s41586-024-08328-6

[19] Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. 2017. LightGBM: a highly efficient gradient boosting decision tree. In *Proceedings of the 31st International Conference on Neural Information Processing Systems* (Long Beach, California, USA) *(NIPS'17)*. Curran Associates Inc., Red Hook, NY, USA, 3149–3157.

[20] Nan Rosemary Ke, Silvia Chiappa, Jane X Wang, Jorg Bornschein, Anirudh Goyal, Melanie Rey, Theophane Weber, Matthew Botvinick, Michael Curtis Mozer, and Danilo Jimenez Rezende. 2023. Learning to Induce Causal Structure. In *International Conference on Learning Representations*. https://openreview.net/forum?id=hp_RwhKDJ5

[21] Jannik Kossen, Neil Band, Clare Lyle, Aidan Gomez, Tom Rainforth, and Yarin Gal. 2021. Self-Attention Between Datapoints: Going Beyond Individual Input-Output Pairs in Deep Learning. In *Advances in Neural Information Processing Systems*, A. Beygelzimer, Y. Dauphin, P. Liang, and J. Wortman Vaughan (Eds.). https://openreview.net/forum?id=wRXzOa2z5T

[22] Kun Kuang, Peng Cui, Susan Athey, Ruoxuan Xiong, and Bo Li. 2018. Stable Prediction across Unknown Environments. In *KDD*. 1617–1626. https://doi.org/10.1145/3219819.3220082

[23] Zhaolong Ling, Kui Yu, Hao Wang, Lin Liu, Wei Ding, and Xindong Wu. 2019. BAMB: A Balanced Markov Blanket Discovery Approach to Feature Selection. *ACM Trans. Intell. Syst. Technol.* 10, 5 (2019), 52:1–52:25. https://doi.org/10.1145/3335676

[24] Lars Lorch, Scott Sussex, Jonas Rothfuss, Andreas Krause, and Bernhard Schölkopf. 2022. Amortized Inference for Causal Structure Learning. In *Advances in Neural Information Processing Systems*, S. Koyejo, S. Mohamed, A. Agarwal, D. Belgrave, K. Cho, and A. Oh (Eds.), Vol. 35. Curran Associates, Inc., 13104–13118. https://proceedings.neurips.cc/paper_files/paper/2022/file/54f7125dee9b8b3dc798bb9a082b09e2-Paper-Conference.pdf

[25] Yuchen Ma, Dennis Frauen, Emil Javurek, and Stefan Feuerriegel. 2026. Foundation Models for Causal Inference via Prior-Data Fitted Networks. In *The Fourteenth International Conference on Learning Representations*. https://openreview.net/forum?id=d2L1ndOKjq

[26] Dimitris Margaritis and Sebastian Thrun. 1999. Bayesian Network Induction via Local Neighborhoods. In *Advances in Neural Information Processing Systems*, S. Solla, T. Leen, and K. Müller (Eds.), Vol. 12. MIT Press. https://proceedings.neurips.cc/paper_files/paper/1999/file/5d79099fcdf499f12b79770834c0164a-Paper.pdf

[27] Samuel Müller, Noah Hollmann, Sebastian Pineda Arango, Josif Grabocka, and Frank Hutter. 2022. Transformers Can Do Bayesian Inference. In *International Conference on Learning Representations*. https://openreview.net/forum?id=KSugKcbNf9

[28] Thomas Nagler. 2023. Statistical foundations of prior-data fitted networks. In *Proceedings of the 40th International Conference on Machine Learning* (Honolulu, Hawaii, USA) *(ICML'23)*. JMLR.org, Article 1066, 17 pages.

[29] Eshaan Nichani, Alex Damian, and Jason D. Lee. 2024. How transformers learn causal structure with gradient descent. In *Proceedings of the 41st International Conference on Machine Learning* (Vienna, Austria) *(ICML'24)*. JMLR.org, Article 1543, 53 pages.

[30] Kirtan Padh, Zhufeng Li, Cecilia Casolo, and Niki Kilbertus. 2025. Your Assumed DAG is Wrong And Here's How To Deal With It. In *Proceedings of the Fourth Conference on Causal Learning and Reasoning (Proceedings of Machine Learning Research, Vol. 275)*, Biwei Huang and Mathias Drton (Eds.). PMLR, 1239–1267. https://proceedings.mlr.press/v275/padh25a.html

[31] Judea Pearl. 2014. *Probabilistic reasoning in intelligent systems: networks of plausible inference*. Elsevier.

[32] Liudmila Prokhorenkova, Gleb Gusev, Aleksandr Vorobev, Anna Veronika Dorogush, and Andrey Gulin. 2018. CatBoost: unbiased boosting with categorical features. In *Proceedings of the 32nd International Conference on Neural Information Processing Systems* (Montréal, Canada) *(NIPS'18)*. Curran Associates Inc., Red Hook, NY, USA, 6639–6649.

[33] Jake Robertson, Arik Reuter, Siyuan Guo, Noah Hollmann, Frank Hutter, and Bernhard Schölkopf. 2025. Do-PFN: In-Context Learning for Causal Effect Estimation. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*. https://openreview.net/forum?id=OaNbl9b56B

[34] Raanan Y. Rohekar, Yaniv Gurwicz, and Shami Nisimov. 2023. Causal interpretation of self-attention in pre-trained transformers. In *Proceedings of the 37th International Conference on Neural Information Processing Systems* (New Orleans, LA, USA) *(NIPS '23)*. Curran Associates Inc., Red Hook, NY, USA, Article 1366, 16 pages.

[35] Zheyan Shen, Peng Cui, Kun Kuang, Bo Li, and Peixuan Chen. 2018. Causally Regularized Learning with Agnostic Data Selection Bias. In *ACM Multimedia*. 411–419. https://doi.org/10.1145/3240508.3240577

[36] Zheyan Shen, Peng Cui, Tong Zhang, and Kun Kuang. 2020. Stable Learning via Sample Reweighting. In *AAAI*. 5692–5699. https://ojs.aaai.org/index.php/AAAI/article/view/6024

[37] Shohei Shimizu, Patrik O. Hoyer, Aapo Hyvärinen, and Antti Kerminen. 2006. A Linear Non-Gaussian Acyclic Model for Causal Discovery. *Journal of Machine Learning Research* 7, 72 (2006), 2003–2030. http://jmlr.org/papers/v7/shimizu06a.html

[38] Peter Spirtes and Clark Glymour. 1991. An Algorithm for Fast Recovery of Sparse Causal Graphs. *Social Science Computer Review* 9, 1 (1991), 62–72. arXiv:https://doi.org/10.1177/089443939100900106 doi:10.1177/089443939100900106

[39] Robert Tibshirani. 1996. Regression Shrinkage and Selection via the Lasso. *Journal of the Royal Statistical Society. Series B (Methodological)* 58, 1 (1996), 267–288. http://www.jstor.org/stable/2346178

[40] I. Tsamardinos, Constantin F. Aliferis, and Alexander R. Statnikov. 2003. Time and sample efficient discovery of Markov blankets and direct causal relations. In *Knowledge Discovery and Data Mining*. https://api.semanticscholar.org/CorpusID:14900819

[41] Haotian Wang, Kun Kuang, Long Lan, Zige Wang, Wanrong Huang, Fei Wu, and Wenjing Yang. 2024. Out-of-Distribution Generalization With Causal Feature Separation. *IEEE Trans. Knowl. Data Eng.* 36, 4 (April 2024), 1758–1772. https://doi.org/10.1109/TKDE.2023.3312255

[42] Hao Wang, Zhaolong Ling, Kui Yu, and Xindong Wu. 2020. Towards efficient and effective discovery of Markov blankets for feature selection. *Inf. Sci.* 509 (2020), 227–242. https://doi.org/10.1016/j.ins.2019.09.010

[43] Xingyu Wu, Bingbing Jiang, Kui Yu, Chunyan Miao, and Huanhuan Chen. 2020. Accurate Markov Boundary Discovery for Causal Feature Selection. *IEEE Trans. Cybern.* 50, 12 (2020), 4983–4996. https://doi.org/10.1109/TCYB.2019.2940509

[44] Yiquan Wu, Yifei Liu, Ziyu Zhao, Weiming Lu, Yating Zhang, Changlong Sun, Fei Wu, and Kun Kuang. 2024. De-biased Attention Supervision for Text Classification with Causality. *Proceedings of the AAAI Conference on Artificial Intelligence* 38, 17 (Mar. 2024), 19279–19287. doi:10.1609/aaai.v38i17.29897

[45] Renzhe Xu, Xingxuan Zhang, Zheyan Shen, Tong Zhang, and Peng Cui. 2022. A Theoretical Analysis on Independence-driven Importance Weighting for Covariate-shift Generalization. In *ICML*. 24803–24829. https://proceedings.mlr.press/v162/xu22o.html

[46] Han-Jia Ye, Si-Yang Liu, and Wei-Lun Chao. 2025. A Closer Look at TabPFN v2: Understanding Its Strengths and Extending Its Capabilities. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*. https://openreview.net/forum?id=lQYNmlTwtc

[47] Kui Yu, Xianjie Guo, Lin Liu, Jiuyong Li, Hao Wang, Zhaolong Ling, and Xindong Wu. 2021. Causality-based Feature Selection: Methods and Evaluations. *ACM Comput. Surv.* 53, 5 (2021), 111:1–111:36. https://doi.org/10.1145/3409382

[48] Kui Yu, Lin Liu, Jiuyong Li, and Huanhuan Chen. 2018. Mining Markov Blankets Without Causal Sufficiency. *IEEE Trans. Neural Networks Learn. Syst.* 29, 12 (2018), 6333–6347. https://doi.org/10.1109/TNNLS.2018.2828982

[49] Xiyuan Zhang, Danielle C. Maddix, Junming Yin, Nick Erickson, Abdul Fatir Ansari, Boran Han, Shuai Zhang, Leman Akoglu, Christos Faloutsos, Michael W. Mahoney, Cuixiong Hu, Huzefa Rangwala, George Karypis, and Bernie Wang. 2025. Mitra: Mixed Synthetic Priors for Enhancing Tabular Foundation Models. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*. https://openreview.net/forum?id=t8YRsWY6HM

[50] Xingxuan Zhang, Gang Ren, Han Yu, Hao Yuan, Hui Wang, Jiansheng Li, Jiayun Wu, Lang Mo, Li Mao, Mingchao Hao, Ningbo Dai, Renzhe Xu, Shuyang Li, Tianyang Zhang, Yue He, Yuanrui Wang, Yunjia Zhang, Zijing Xu, Dongzhe Li, Fang Gao, Hao Zou, Jiandong Liu, Jiashuo Liu, Jiawei Xu, Kaijie Cheng, Kehan Li, Linjun Zhou, Qing Li, Shaohua Fan, Xiaoyu Lin, Xinyan Han, Xuanyue Li, Yan Lu, Yuan Xue, Yuanyuan Jiang, Zimu Wang, Zhenlei Wang, and Peng Cui. 2025. LimiX: Unleashing Structured-Data Modeling Capability for Generalist Intelligence. *CoRR* abs/2509.03505 (September 2025). https://doi.org/10.48550/arXiv.2509.03505

A Implementation Details

In this section, we provide details regarding the architecture and optimization strategy of StablePFN to ensure reproducibility.

A.1 Model Architecture

StablePFN is built upon the Dual-Axis Attention Block (DAB) , which captures complex tabular dependencies by alternating between **Feature Attention** (across columns) and **Sample Attention** (across rows). The architecture follows a multi-task shared-bottom design:

- **Shared Encoder**: To extract robust global representations $H \in \mathbb{R}^{N\times(F+1)\times E}$, we employ $L_{sh} = 4$ DAB layers.
- **Functional Branches**: The model splits into three parallel pipelines to execute the "Decouple-Discover-Predict" strategy. The Decoupling, Discovery, and Prediction branches are configured with $L_{de} = 2$, $L_{di} = 4$, and $L_{pr} = 6$ DAB layers respectively to balance computational efficiency with representational power.
- **Embedding & Feature Identity**: We set the embedding dimension $E = 128$ with $H_{head} = 4$ attention heads. Since Transformers are permutation-invariant, we add Random Feature ID Embeddings to the input representations to preserve the unique identity of each column across different tabular tasks.
- **Branch-Specific Heads**:
  - *Feature-to-Weight Aggregation Head*: Utilizes a query token $\mathbf{q} \in \mathbb{R}^F$ and a cross-attention mechanism to aggregate feature-wise information into sample weights $\hat{w}$.
  - *Weighted MB Head*: Performs a signed logarithmic transformation on the weighted cross-similarity to produce the final MB probability scores $\mathbf{s} \in [0, 1]^F$.
  - *Prediction Head*: Following the PFN standard, we approximate the Posterior Predictive Distribution (PPD) using a categorical distribution over discretized bins. The continuous target space is discretized into $B = 100$ intervals, and we incorporate a log-space linear penalty term to handle out-of-distribution targets.

A.2 Training Protocol and Optimization

StablePFN is trained in an multi-task learning fashion. The joint loss function balances structural discovery and predictive accuracy:

$$\mathcal{L}_{total} = \lambda_1 \mathcal{L}_w + \lambda_2 \mathcal{L}_{disc} + \lambda_3 \mathcal{L}_{pred} \tag{16}$$

where we set $\lambda_1 = 0.1$ to regularize the decoupling weights, and $\lambda_2 = \lambda_3 = 1.0$ to emphasize both MB discovery and PPD fitting, and set the internal regularization weight $\lambda_{\text{reg}} = 10^{-3}$. We use the `AdamW` optimizer with a learning rate of 1×10^{-4} and a cosine learning rate scheduler. To prevent gradients from the Discovery Branch from destabilizing the Decoupling Branch, a *stop-gradient* operator $sg[\hat{w}]$ is applied during the computation of the MB Association.

B Additional Model Analyses

In this section, we provide additional empirical analyses of the intrinsic properties of StablePFN, including its sensitivity to the masking threshold, the necessity of the decoupling component, and its computational complexity.

B.1 Threshold Sensitivity for Hard MB Masking

In Section 4.3, we applied a Hard MB Mask using a default threshold of $\tau = 0.5$. Table 3 presents the threshold sensitivity evaluated on the real-world benchmark from Section 5.3 under $\rho = 0.5$.

Performance remains highly robust for $\tau \geq 0.5$. Conversely, lower thresholds ($\tau < 0.5$) cause severe performance drops (negative R^2) as they fail to strictly filter out spurious features. We adopt hard masking over soft masking because soft attention mechanisms may still permit residual spurious signals to influence inference, whereas hard masking enforces structural constraints explicitly.

B.2 Ablation on the Decoupling Mechanism

Decoupling is the core of the Stable Learning framework and serves as the theoretical foundation for identifiability. To isolate its contribution, we conducted an ablation study comparing a variant retrained completely from scratch under the same protocol but lacking the Decoupling Branch and its corresponding loss ($\mathcal{L}_w$) on the real-world benchmark from Section 5.3 under $\rho = 0.5$.

As shown in Table 4, without the decoupling mechanism, the model's stable prediction performance deteriorates substantially to near-zero or negative R^2. This drop occurs because similarity in the original observational space is directly distorted by spurious associations, whereas mapping into the decoupled space effectively mitigates this issue.

Table 3: Threshold sensitivity evaluated on the real-world benchmark ($\rho = 0.5$).

τ	Mean R^2
0.1	-0.318 ± 0.53
0.3	-0.301 ± 0.54
0.5	0.142 ± 0.34
0.7	0.156 ± 0.32
0.9	0.122 ± 0.29

Table 4: Ablation study on the decoupling mechanism (Mean R^2). "w/o Decoup." denotes the model without the decoupling mechanism.

Model	$\rho = 0.3$	$\rho = 0.5$	$\rho = 0.7$
StablePFN	0.135 ± 0.43	0.142 ± 0.34	0.082 ± 0.31
w/o Decoup.	-0.048 ± 0.53	-0.024 ± 0.41	-0.130 ± 0.36

Table 5: Computational cost and complexity evaluated on 21 datasets.

Model	Params	VRAM	Time
StablePFN	4.5M	4.5GB	458ms
TabPFN	10M	0.9GB	335ms
LimiX	16M	2.1GB	223ms

B.3 Computational Cost and Complexity

StablePFN shares the same $O(N^2 + F^2)$ asymptotic complexity as standard DAB-based models. In Table 5, we summarize the empirical cost evaluated on 21 real-world datasets from Section 5.3 using a single NVIDIA RTX 4090 GPU.

The computational overhead represents a moderate constant factor. This is primarily attributed to two architectural factors: (1) the parallel execution of the "Decouple-Discover-Predict" branches, and (2) the absence of feature grouping operations in our model compared to highly optimized baselines. Crucially, while existing models (e.g., TabPFN v2.5 and LimiX) achieve superior memory and speed efficiency through structural feature grouping, such an operation may be undesirable in causal-aware scenarios. Causal and MB discovery rely on variable-level information to recover structural relationships. Grouping variables may obscure token-level feature boundaries and make recovering structural relationships substantially more difficult. Therefore, StablePFN intentionally retains fine-grained feature attention to better preserve causal structure, accepting a modest efficiency trade-off.

C Additional Experimental Results

Due to strict space limitations, the comprehensive empirical results and extended analyses have been omitted from this camera-ready version. The omitted content primarily includes:

- Granular MB Discovery Metrics: Detailed evaluations, including F1-score, Accuracy, Precision, and Recall, across varying synthetic graph complexities.
- Comprehensive Real-world Evaluations: The complete tabular results reporting Mean Squared Error (MSE) and R^2 scores for all 21 real-world datasets.

For the unabridged version of this manuscript containing the complete Appendix C, alongside the code and model of StablePFN, please refer to our official GitHub repository: https://github.com/GuanZhengkang/StablePFN

D LLM-Assisted Writing Disclosure

During the preparation of this manuscript, the authors utilized LLMs solely for the purpose of language editing, grammatical correction, and polishing to enhance the readability of the text. The authors retain full responsibility for the integrity and originality of the entire manuscript.